Introduction to Virtual/Augmented Reality and Telepresence

Assignment 4
Hummingbirds Creature Using Reinforced Learning

In this assignment, you will learn how to create intelligent flying hummingbirds that can navigate to flowers, dip their beaks in, and drink nectar. These hummingbirds have six degrees of freedom, meaning they can fly and turn in any direction to find targets. They have more complicated controls and their flight paths cannot be solved with traditional navigation systems. You will learn how to craft a training environment and train neural networks to perform this challenging task, then compete against the birds yourself in a simple mini game.

There's no question that Reinforcement Learning is a complicated subject, but ML-Agents leaves the hard parts to the researchers so that you can focus on solving problems. By the end of the assignment, you'll have a good understanding of how you can create intelligent agents and integrate them with your own Unity games or simulation projects. **Make sure you also use the notes in the correspond lecture to get a better understanding of the how the code works.**

The Unity 3D assignment is located at https://learn.unity.com/course/ml-agents-hummingbirds?version=2019.3

The making will be based on the following:

- The project should work as described in the tutorial: 70%
- Improved functionalities: Better training 10%, improved game interface 10%, any new additions 10%

Send the working game (as a video capture) and a link to the Unity 3D project at VRARMM806.2025@gmail.com on December 5.